α-VALUATIONS OF C_4-BOUQUETS

Abstract. For $n \geq 1$, let B_n be the bouquet obtained by identifying one vertex in each of n copies of C_4. We prove that B_n admits an α-valuation if and only if $n \leq 4$. The existence half is known: for $2 \leq n \leq 4$ it is the case $m = 4$ of Theorem 3.8 of Figueroa-Centeno, Ichishima, and Muntaner-Batle, and $B_1 = C_4$ is an α-graph by Rosa's classification of cycles; what is new here is that no $n \geq 5$ admits one. In particular, B_5, with 16 vertices and 20 edges, is a bipartite counterexample to the sufficiency direction of Conjecture 3.10 of the same paper. Thus the conjecture fails even for $m = 4$.

1. The result

A graceful labeling of a graph with q edges is an injection $f\colon V(G) \to \{0, 1, \ldots, q\}$ whose induced edge differences are $1, 2, \ldots, q$. It is an α-valuation if there is an integer k such that every edge has one endpoint labeled at most k and the other labeled greater than k.

Figueroa-Centeno, Ichishima, and Muntaner-Batle conjectured that the one-point union $\mathrm{Amal}(C_m, v, n)$ admits an α-valuation if and only if $mn \equiv 0 \pmod 4$ [1, Conjecture 3.10]. The twenty-eighth edition of Gallian's survey records the same conjecture [2, p. 53]. Because every graph admitting an α-valuation is bipartite, odd m with $4 \mid n$ are immediate exceptions to the statement as printed. The nontrivial question is whether the divisibility condition is sufficient for even m. The theorem below answers this negatively already for $m = 4$ and determines all n in that case.

The positive part of that determination is due to the authors of the conjecture: if $m \equiv 0 \pmod 4$ and $n \in \{2, 3, 4\}$, then $\mathrm{Amal}(C_m, v, n)$ admits an α-valuation [1, Theorem 3.8]. (Theorem 3.8 is stated there for $n = 1, 2, 3, 4$, although $\mathrm{Amal}(G, v, n)$ is defined in [1] only for $n \geq 2$; the remaining case is $\mathrm{Amal}(C_4, v, 1) = C_4$, an α-graph because $4 \equiv 0 \pmod 4$ [3].) What is new below is the nonexistence for every $n \geq 5$.

For $n \geq 1$, write

$$B_n = \mathrm{Amal}(C_4, v, n),$$

and denote its common vertex by h. Label the remaining vertices so that its ith square is

$$h x_i y_i z_i h, \qquad 1 \leq i \leq n.$$

Theorem 1. *The graph B_n admits an α-valuation if and only if $n \leq 4$.*

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Proof. For $1 \le n \le 4$ an α-valuation exists by [1, Theorem 3.8] (case $m = 4$, $n \ge 2$) and by [3] (case $n = 1$). The following table records explicit witnesses, so that the argument here is self-contained and the positive direction can be checked by hand. The triples are $(f(x_i), f(y_i), f(z_i))$, in increasing order of i, and k is the characteristic.

n	$f(h)$	k	$(f(x_i), f(y_i), f(z_i))_{i=1}^n$
1	0	2	$(4, 2, 3)$
2	0	2	$(3, 2, 4),\ (6, 1, 8)$
3	1	3	$(6, 0, 12),\ (4, 3, 5),\ (9, 2, 11)$
4	2	4	$(7, 1, 9),\ (15, 0, 16),\ (5, 4, 6),\ (12, 3, 14)$

In each row the labels are injective, the two partite sets are separated at k, and the edge differences are $1, 2, \ldots, 4n$.

The remainder of the proof is the other direction. Suppose that B_n has an α-valuation f. Its bipartition is

$$A = \{h, y_1, \ldots, y_n\}, \qquad B = \{x_i, z_i : 1 \le i \le n\}.$$

Since B_n is connected, the lower and upper label classes are its two partite sets. Replacing f by $4n - f$ if necessary, assume that A is the lower class. Set

$$S = f(A), \qquad T = f(B), \qquad a = f(h), \qquad U = S \setminus \{a\}.$$

Thus every element of S is smaller than every element of T. For $t \in T$, let $g(t)$ be the label of the vertex y_i in the same square as the vertex of B labeled t. Then

$$g^{-1}(f(y_i)) = \{f(x_i), f(z_i)\},$$

so every fiber of $g\colon T \to U$ has size two.

The edge differences decompose as

(1)
$$\Delta_h = \{t - a : t \in T\}, \qquad \Delta_y = \{t - g(t) : t \in T\}, \qquad \Delta_h \dot\cup \Delta_y = \{1, \ldots, 4n\}.$$

The difference $4n$ forces the labels 0 and $4n$ to occur. Hence $0 \in S$ and $4n \in T$.

We first show that

$$a \le 2. \tag{2}$$

This is immediate if $a = 0$. If $a > 0$, then $\max \Delta_h = 4n - a$, so every element of

$$I = \{4n - a + 1, \ldots, 4n\}$$

lies in Δ_y. For each $d \in I$, let t_d be the unique $t \in T$ with $d = t - g(t)$; it is unique because the differences in (1) are pairwise distinct. We call t_d the label *serving* d. Since $t_d = d + g(t_d)$, we have $t_d \in I$. The labels t_d are distinct, and I has a elements; therefore $\{t_d : d \in I\} = I$. Consequently,

$$\sum_{d \in I} g(t_d) = \sum_{d \in I} t_d - \sum_{d \in I} d = 0.$$

Thus every $g(t_d)$ equals 0, so $0 \in U$. Since $|g^{-1}(0)| = 2$, we obtain $a = |I| \leq 2$.

Let

$$M = \max S, \qquad \tau = \min T.$$

We next show that

$$M - a \leq 2. \tag{3}$$

Since $\min \Delta_h = \tau - a$, every integer $1, \dots, \tau - a - 1$ lies in Δ_y. If $d = t_d - g(t_d)$ is one of these differences, then

$$\tau \leq t_d = d + g(t_d) \leq \tau - a - 1 + M.$$

The serving labels t_d are distinct, while this interval contains $M - a$ integers. Hence

$$\tau - a - 1 \leq M - a$$

(if $\tau - a - 1 \leq 0$ there are no such differences and the inequality is trivial, since $a \in S$ gives $M \geq a$). Because $\tau > M$, it follows that $\tau = M + 1$.

Put $r = M - a$. If $r = 0$, (3) is immediate. If $r > 0$, the small differences are $1, \dots, r$, and their serving labels exhaust $M + 1, \dots, M + r$. Therefore

$$\sum_{d=1}^{r} g(t_d) = \sum_{j=1}^{r} (M + j) - \sum_{d=1}^{r} d = rM.$$

Since $r > 0$, we have $M \in U$. Every summand is at most M, so all of them equal M. Hence $r \leq |g^{-1}(M)| = 2$, proving (3).

By (2) and (3), $M \leq 4$. The set S consists of $n + 1$ distinct labels in $\{0, 1, \dots, M\}$, so $n \leq M \leq 4$. $\square$

Both inequalities are attained by the $n = 4$ row of the table, so the proof is self-certifying at its own boundary: there $a = 2$, $M = 4$, $\tau = M + 1 = 5$, and the two fibers named in the proof are $g^{-1}(0) = \{15, 16\}$ and $g^{-1}(4) = \{5, 6\}$. Thus (2) and (3) are simultaneously saturated by an exhibited labeling, and the chain $n \leq M = a + (M - a) \leq 2 + 2$ cannot be improved at either step.

Corollary 2. *For every $n \geq 5$, the bipartite graph B_n has $4n$ edges and admits no α-valuation. Hence B_5 refutes the conjectured sufficiency condition within the even-cycle subfamily of Conjecture 3.10 in* [1].

Proof. For $m = 4$, the conjectured divisibility condition holds for every n, whereas Theorem 1 excludes every $n \geq 5$. $\square$

Status. What is refuted is Conjecture 3.10 of [1] as stated, and what is in addition settled is its $m = 4$ column, completely. The existence half of Theorem 1 is not ours: it is [1, Theorem 3.8] for $n = 2, 3, 4$ and [3] for $n = 1$, and the table above only exhibits witnesses for it. Nothing else is claimed. The nonexistence argument uses $m = 4$ essentially — each vertex of B is adjacent only to h and to the y of its own square, which fails for every longer cycle — so the columns $m \equiv 0 \pmod 4$ with $m \geq 8$, and $m \equiv 2 \pmod 4$ with n even and $n \geq 6$, are untouched here. In the former columns the cases $n \leq 4$ are already settled positively by [1, Theorem 3.8], so what remains open there is the range $n \geq 5$. No minimality claim is made: B_5 is exhibited as a bipartite counterexample, not asserted to be the smallest counterexample of any kind.

The obstruction is specific to the α-property and not to gracefulness: $\mathrm{Amal}(C_4, v, n)$ is graceful for every n, by a result of Shee recorded in [2, p. 19]. So the conjecture of Koh, Rogers, Lee, and Toh that $\mathrm{Amal}(C_m, v, n)$ is graceful if and only if $mn \equiv 0$ or $3 \pmod 4$, from which Conjecture 3.10 is descended, is not affected by anything above.

References

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